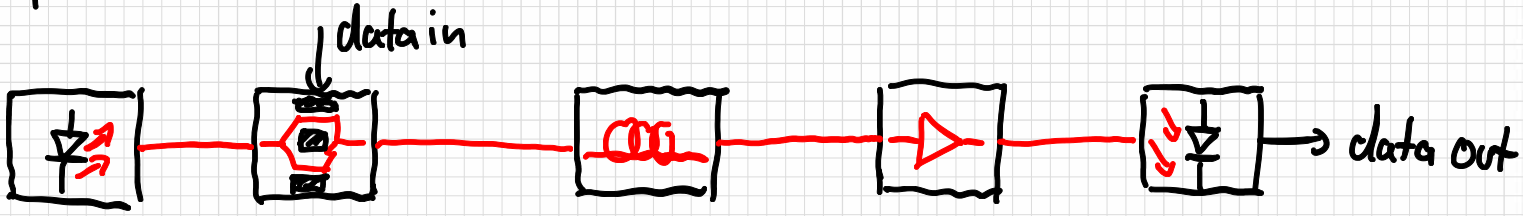
22.10.2024

Introduction

Optical communications channel:



Laser	Mach-Zehnder modulator	Fiber	Optical amplifier	Photodiode
Particles	Electromagnetic waves	Ray optics	Particles	Particles
		Electromagnetic waves	Electromagnetic waves	Ray optics

Three different posibilities to describe the physics of light:

① Ray optics

A ray represents the direction of propagation of an electromagnetic wave. It correlates with the wave vector $\vec{k}$.

$$\vec{k} = \begin{pmatrix} k_x \\ k_y \\ k_z \end{pmatrix} \Rightarrow k = |\vec{k}| = \sqrt{k_x^2 + k_y^2 + k_z^2} = n \cdot k_0 = n \cdot \frac{2\pi}{\lambda} = n \cdot \frac{2\pi f}{c_0}$$

A ray is a straight line parallel to the wave vector.

- Easy calculation
- Purely geometric calculation of reflection and refraction.

② Electromagnetic waves

Maxwell's equations with $\mu_r = 1$, $\rho = 0$ in Cartesian coordinates $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\text{curl } \vec{E}(\vec{r}) = -j\omega\mu_0 \vec{H}(\vec{r})$$

$$\text{curl } \vec{H}(\vec{r}) = \vec{J}(\vec{r}) + j\omega\varepsilon_0\varepsilon_r \vec{E}(\vec{r}) \quad \text{with } \vec{J}(\vec{r}) = G \cdot \vec{E}(\vec{r})$$

↳ electrical conductivity

$$\text{div } \vec{E}(\vec{r}) = 0$$

$$\text{div } \vec{H}(\vec{r}) = 0$$

Solution: $\vec{E}(\vec{r},t) = \vec{E}_0 e^{-j\underline{\vec{k}}\vec{r}} e^{j\omega t}$, $\vec{H}(\vec{r},t) = \vec{H}_0 e^{-j\underline{\vec{k}}\vec{r}} e^{j\omega t}$

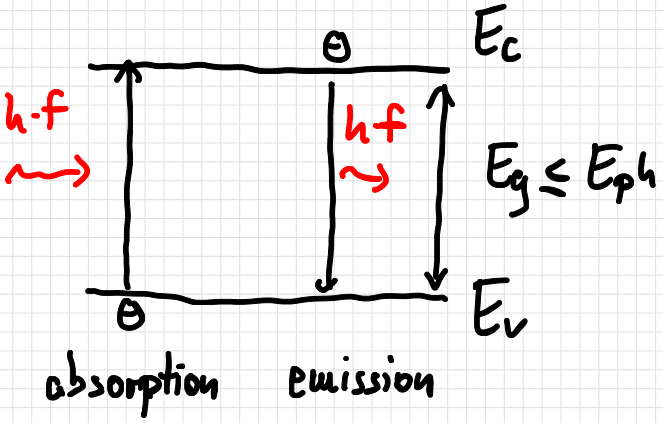
with $|\underline{\vec{k}}| = \underline{k} = \omega\sqrt{\left(\varepsilon_r - j\frac{G}{\omega\varepsilon_0}\right)\varepsilon_0\mu_0}$

→ Frequency, wavelength, velocity

③ Particles

Photon energy: $E_{ph} = h \cdot f = h \cdot \frac{c_0}{\lambda}$ with $h = 6.626 \cdot 10^{-34}\ \text{Ws}^2$

... Planck's constant



Problem 1

The wavelength describes the distance an electromagnetic wave propagates in a medium during one period of its oscillating frequency.

$$\lambda = \frac{c_0}{f} = \frac{c_0}{1/T} = c_0 T$$

With $\omega = 1.216 \cdot 10^{15} \frac{rad}{s}$:

$$\lambda = \frac{c_0}{f} = \frac{c_0}{\frac{\omega}{2\pi}} = \frac{2\pi c_0}{\omega} = \frac{2\pi \cdot 3 \cdot 10^8 \frac{m}{s}}{1.216 \cdot 10^{15} \frac{rad}{s}} = 1.55\,\mu m$$

$$\lambda_{H2O} = \frac{2\pi v_{ph,H2O}}{\omega} = \frac{2\pi \, c_0 / n_{H2O}}{\omega} = \frac{1}{n_{H2O}} \frac{2\pi c_0}{\omega} = \frac{\lambda}{n_{H2O}} = \frac{1.55\,\mu m}{1.33} = 1.165\,\mu m$$

$$\lambda_{SiO_2} = \frac{\lambda}{n_{SiO_2}} = \frac{1.55\,\mu m}{1.44} = 1.076\,\mu m$$

$$\lambda_{GaAs} = \frac{\lambda}{n_{GaAs}} = \frac{1.55\,\mu m}{3.6} = 0.431\,\mu m$$

Problem 2

a) $\underline{\vec{E}}(z,t) = \begin{pmatrix} \underline{E_x} e^{j(\omega t - \underline{k} z)} \\ 0 \\ 0 \end{pmatrix} = \vec{e}_x \, \underline{E_x} e^{j(\omega t - \underline{k} z)}$

$$\nabla^2 \underline{\vec{E}}(z,t) + \mu_0 \varepsilon_0 \underline{\varepsilon_r} \omega^2 \underline{\vec{E}}(z,t) = 0$$

$$\Rightarrow \quad \nabla^2 \underline{\vec{E}}(z,t) = -\mu_0 \varepsilon_0 \underline{\varepsilon_r} \omega^2 \underline{\vec{E}}(z,t)$$

$\nabla^2 \hat{=} \Delta \hat{=} \frac{\partial^2}{\partial x^2} \underline{\vec{E}} + \frac{\partial^2}{\partial y^2} \underline{\vec{E}} + \frac{\partial^2}{\partial z^2} \underline{\vec{E}}$:

$$\nabla^2 \underline{\vec{E}}(z,t) = \vec{e}_x \nabla^2 \left(\underline{E_x} e^{j(\omega t - \underline{k} z)} \right) = \vec{e}_x \frac{\partial^2}{\partial z^2} \underline{E_x} e^{j(\omega t - \underline{k} z)} =$$

$$= \vec{e}_x (-j\underline{k})^2 \underline{E_x} e^{j(\omega t - \underline{k} z)}$$

$$\vec{e}_x (-j\underline{k})^2 \underline{E_x} e^{j(\omega t - \underline{k} z)} = -\mu_0 \varepsilon_0 \underline{\varepsilon_r} \omega^2 \vec{e}_x \underline{E_x} e^{j(\omega t - \underline{k} z)}$$

$$(-j\underline{k})^2 = -\mu_0 \varepsilon_0 \underline{\varepsilon_r} \omega^2$$

$$\underline{k}^2 = \mu_0 \varepsilon_0 \underline{\varepsilon_r} \omega^2$$

$$\underline{k} = \omega \sqrt{\mu_0 \varepsilon_0 \underline{\varepsilon_r}} \quad \text{with} \quad \underline{\varepsilon_r} = \varepsilon_r' - j\varepsilon_r'' = \varepsilon_r' - j\frac{\sigma}{\omega \varepsilon_0}$$

σ ... electric conductivity $\left[\frac{S}{m} = \frac{1}{\Omega m} \right]$

b) $\underline{k} = \beta - j\alpha$

β ↙ phase coefficient; α ↘ absorption coefficient

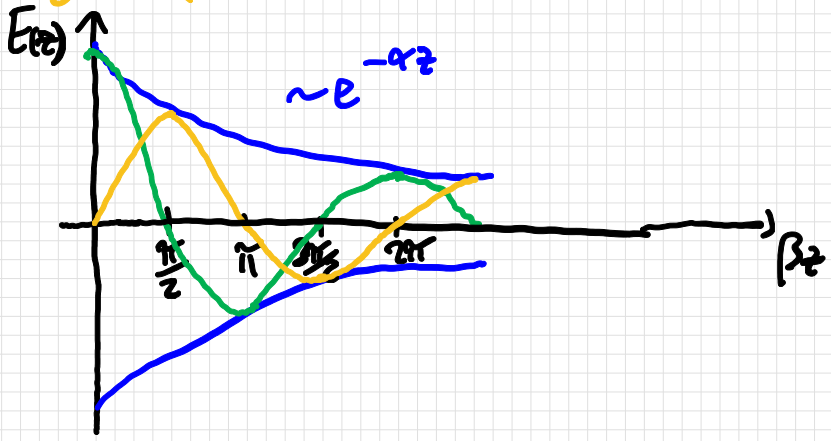
$$\underline{\vec{E}}(z,t) = \vec{e}_x \underline{E_x} e^{j(\omega t - \underline{k} z)} = \vec{e}_x \underline{E_x} e^{j\omega t} e^{-j(\beta - j\alpha) z} = \vec{e}_x \underline{E_x} e^{j\omega t} e^{-j\beta z} e^{-\alpha z} =$$

$$= \vec{e}_x \underline{E_x} e^{-\alpha z} \left(\cos(\omega t - \beta z) + j \sin(\omega t - \beta z) \right)$$

$$\mathrm{Re}\{\underline{\vec{E}}(z,t)\}$$

$\omega t = 0$: $\mathrm{Re}\{\underline{\vec{E}}(z,t)\} = \vec{e}_x E_x e^{-\alpha z}\cos(-\beta z) = \vec{e}_x E_x e^{-\alpha z}\cos(\beta z)$

$\omega t = \frac{\pi}{2}$: $\mathrm{Re}\{\underline{\vec{E}}(z,t)\} = \vec{e}_x E_x e^{-\alpha z}\cos(\frac{\pi}{2} - \beta z) = \vec{e}_x E_x e^{-\alpha z}\sin(\beta z)$



c) $\underline{Z} = \sqrt{\frac{\mu_0\mu_r}{\varepsilon_0\underline{\varepsilon}_r}} = \sqrt{\frac{\mu_0}{\varepsilon_0\underline{\varepsilon}_r}} = \sqrt{\frac{\mu_0}{\varepsilon_0}}\frac{1}{\sqrt{\underline{\varepsilon}_r}} = Z_F\frac{1}{\sqrt{\underline{\varepsilon}_r}}$ with $Z_F = 120\pi\,\Omega \approx 377\,\Omega$

free wave characteristic impedance

$\sqrt{\underline{\varepsilon}_r} = \sqrt{\varepsilon_r' - j\varepsilon_r''} = n' - jn'' = n - j\kappa = |n - j\kappa| e^{j\tan^{-1}(-\frac{\kappa}{n})} = \sqrt{n^2+\kappa^2}\, e^{j\varphi_0}$

refractive index; extinction coefficient

$n \neq \sqrt{\varepsilon_r'}$, $\kappa \neq \sqrt{\varepsilon_r''}$

$\Rightarrow \underline{Z} = Z_F\frac{1}{\sqrt{n^2+\kappa^2}} e^{-j\varphi_0} = |\underline{Z}| e^{-j\varphi_0}$

$\mathrm{curl}\,\vec{E} = -j\omega\mu_0\vec{H} \Rightarrow \vec{H} = -\frac{1}{j\omega\mu_0}\mathrm{curl}\,\vec{E}$

$$\vec{H}(z,t) = -\frac{1}{j\omega\mu_0}\begin{pmatrix}\frac{\partial}{\partial x}\\ \frac{\partial}{\partial y}\\ \frac{\partial}{\partial z}\end{pmatrix} \times \begin{pmatrix} E_x e^{j(\omega t - \underline{k}z)}\\ 0\\ 0\end{pmatrix} = -\frac{1}{j\omega\mu_0}\begin{pmatrix}0\\ \frac{\partial}{\partial z}E_x e^{j(\omega t-\underline{k}z)}\\ 0\end{pmatrix} =$$

$$= \vec{e}_y\left(-\frac{1}{j\omega\mu_0}\right)\frac{\partial}{\partial z}E_x e^{j(\omega t-\underline{k}z)} = \vec{e}_y\underbrace{\frac{-j\underline{k}}{-j\omega\mu_0}E_x e^{j(\omega t-\underline{k}z)}}_{H_y}$$

$$H_y = \frac{k}{\omega\mu_0} E_x = \frac{\omega\sqrt{\mu_0\varepsilon_0\varepsilon_r}}{\omega\mu_0} E_x = \sqrt{\frac{\varepsilon_0\varepsilon_r}{\mu_0}} E_x = \frac{E_x}{\underline{Z}} = \frac{E_x}{|\underline{Z}|} e^{j\varphi_0}$$

$$\vec{\underline{E}}(z,t) = \vec{e}_x \, E_x \, e^{j(\omega t - \underline{k}z)} = \vec{e}_x \, E_x \, e^{-\alpha z} \, e^{j\omega t} \, e^{-j\beta z}$$

$$\vec{\underline{H}}(z,t) = \vec{e}_y \, \frac{E_x}{|\underline{Z}|} \, e^{j(\omega t - \underline{k}z + \varphi_0)} = \vec{e}_y \, \frac{E_x}{|\underline{Z}|} \, e^{-\alpha z} \, e^{j\omega t} \, e^{-j\beta z}$$

